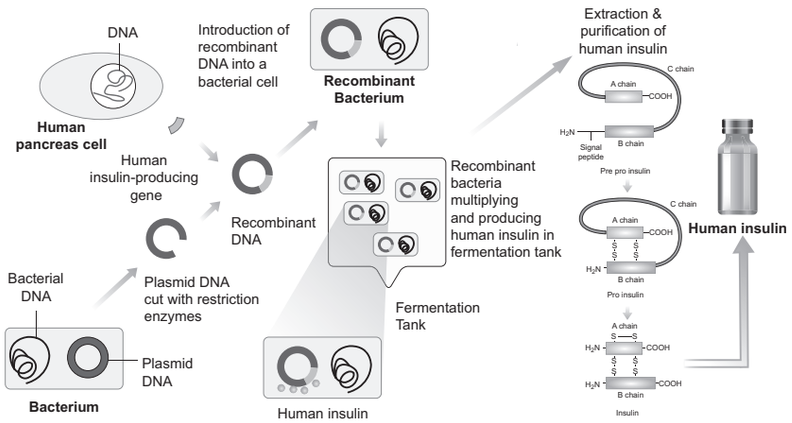
## 16. HUMAN INSULIN PRODUCTION- FLOWCHART

### IDENTIFICATION

The given picture is identified as the flow chart of Human Insulin Production.

### COMMENTS




1. Production of insulin by recombinant DNA technology started in the late 1970s.
2. This technique involved the insertion of human insulin gene on the plasmids of E.coli.
3. The inserted gene synthesizes the polypeptide chains A and B segments linked by a third chain(C) as a precursor called Pre-Pro insulin.
4. The linking C chain is excised, leaving, A and B polypeptide chains.
5. Insulin was the first ever pharmaceutical product of rDNA technology, administered to humans.

# C - GENETICS

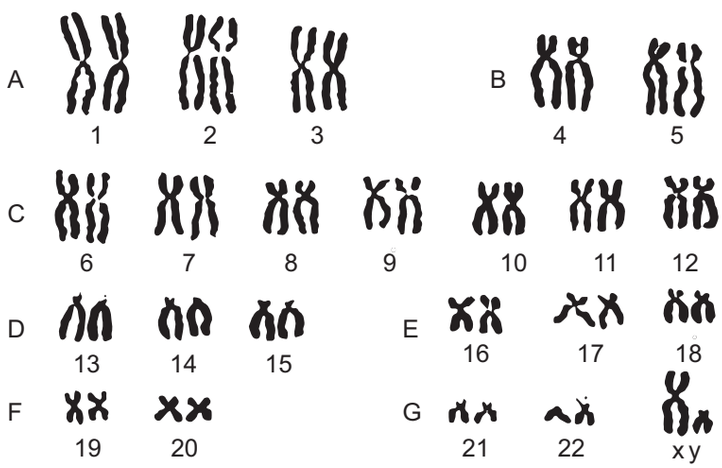
## 17. NORMAL HUMAN KARYOTYPE

### IDENTIFICATION

The given photograph is identified as normal karyotype of human beings.

### COMMENTS

1. Karyotyping is a technique through which a complete set of chromosomes are separated from a cell and are arranged in pairs.
2. A diagrammatic representation of chromosomes is called an idiogram.
3. There are 22 pairs of autosomes and a pair of allosomes ( XX- female, XY – male) arranged based on their size, shape, banding pattern and position of centromere.




4. It helps in gender identification and to detect genetic diseases.



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